13.5 SUPERPOSITION OF TWO SINUSOIDAL WAVES

Let us consider the superposition of two sinusoidal waves (having the same frequency) at a particular point. Let

$$\left.\begin{aligned} x_1(t) &= a_1 \cos(\omega t + \theta_1) \\ x_2(t) &= a_2 \cos(\omega t + \theta_2) \end{aligned}\right\} \quad \text{and} \quad \textbf{(13.20)}$$

represent the displacements produced by each of the disturbances: we are assuming that the displacements are in the same direction*, however, they may have different amplitudes and different initial phases. In Sec. 17.5 now, according to the superposition principle the resultant displacement $x(t)$ would be given by

$$\begin{aligned} x(t) &= x_1(t) + x_2(t) \\ &= a_1 \cos(\omega t + \theta_1) + a_2 \cos(\omega t + \theta_2) \end{aligned} \quad \textbf{(13.21)}$$

which can be written in the form

$$x(t) = a \cos(\omega t + \theta) \quad \textbf{(13.22)}$$

where

$$a \cos\theta = a_1 \cos\theta_1 + a_2 \cos\theta_2 \quad \textbf{(13.23)}$$

and

$$a \sin\theta = a_1 \sin\theta_1 + a_2 \sin\theta_2 \quad \textbf{(13.24)}$$

Thus, the resultant disturbance is also simple harmonic in character having the same frequency but different amplitude and different initial phase. If we square and add Eqs. (13.23) and (13.24), we would obtain

$$a = [a_1^2 + a_2^2 + 2a_1a_2 \cos(\theta_1 - \theta_2)]^{1/2} \quad \textbf{(13.25)}$$

Further

$$\tan\theta = \frac{a_1 \sin\theta_1 + a_2 \sin\theta_2}{a_1 \cos\theta_1 + a_2 \cos\theta_2} \quad \textbf{(13.26)}$$

Though angle θ is not uniquely determined from Eq. (13.26); however, if we assume a to be always positive, then cos θ and sin θ can be determined from Eqs. (13.23) and (13.24) which will uniquely determine θ. From Eq. (13.25) we find that if

$$\theta_1 \sim \theta_2 = 0, 2\pi, 4\pi, \ldots \quad \textbf{(13.27)}$$

then

$$a = a_1 + a_2 \quad \textbf{(13.28)}$$

Thus, if the two displacements are in phase, then the resultant amplitude will be the sum of the two amplitudes; this is known as **constructive interference**. Similarly, if

$$\theta_1 \sim \theta_2 = \pi, 3\pi, 5\pi, \ldots \quad \textbf{(13.29)}$$

then

$$a = a_1 \sim a_2 \quad \textbf{(13.30)}$$

and the resultant amplitude is the difference of the two amplitudes. This is known as **destructive interference**. If we refer to Fig. 13.2, then we can see that constructive interference occurs at $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots$ (i.e., at the points $P', Q', R', \ldots$) and destructive interference occurs at $x = 0, \lambda/2, \lambda, 3\lambda/2, \ldots$ (i.e., at points $A, P, Q, R, \ldots$). It may be mentioned that when constructive and destructive interferences occur, there is no violation of the principle of conservation of energy; the energy is just redistributed.

In general, if we have n displacements

$$\left.\begin{aligned} x_1 &= a_1 \cos(\omega t + \theta_1) \\ x_2 &= a_2 \cos(\omega t + \theta_2) \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ x_n &= a_n \cos(\omega t + \theta_n) \end{aligned}\right\} \quad \textbf{(13.31)}$$

then

$$x = x_1 + x_2 + \ldots + x_n = a \cos(\omega t + \theta) \quad \textbf{(13.32)}$$

where

$$a \cos\theta = a_1 \cos\theta_1 + \ldots + a_n \cos\theta_n \quad \textbf{(13.33)}$$

and

$$a \sin\theta = a_1 \sin\theta_1 + \ldots + a_n \sin\theta_n \quad \textbf{(13.34)}$$

13.6 THE GRAPHICAL METHOD FOR STUDYING SUPERPOSITION OF SINUSOIDAL WAVES

In this section, we will discuss the graphical method for adding displacements of the same frequency. This method is particularly useful when we have a large number of superposing waves as it indeed happens when we consider the phenomenon of diffraction.

* In Sec. 17.5, we will consider the superposition of waves having nearly equal frequencies which leads to the phenomenon of beats. Indeed in Sec. 13.2, while discussing stationary waves on a string, we had, at a particular value of x, two sinusoidal waves of the same frequency (but having different initial phases) superposing on each other. However, in general, one could have superposition of displacements which are in different directions; for example, the superposition of two linearly polarized waves to produce a circularly polarized wave (see Sec. 22.4).

Let us first try to obtain the resultant of the two displacements given by Eq. (13.20) using the graphical method. We draw a circle of radius a_1 and let the point P on the circle be such that OP makes an angle θ_1 with the x-axis* (see Fig. 13.8). We next draw a circle of radius a_2 and let the point Q on the circle be such that OQ makes an angle θ_2 with the x-axis. We use the law of parallelograms to find the resultant $\overrightarrow{OR}$ of the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$. The length of the vector $\overrightarrow{OR}$ will represent the amplitude of the resultant displacement and if θ is the angle that OR makes with the x-axis, then the initial phase of the resultant will be θ. This can be easily seen by noting that

$$OR \cos \theta = OP \cos \theta_1 + PR \cos \theta_2$$
$$= a_1 \cos \theta_1 + a_2 \cos \theta_2 \qquad \textbf{(13.35)}$$

Similarly,

$$OR \sin \theta = a_1 \sin \theta_1 + a_2 \sin \theta_2 \qquad \textbf{(13.36)}$$

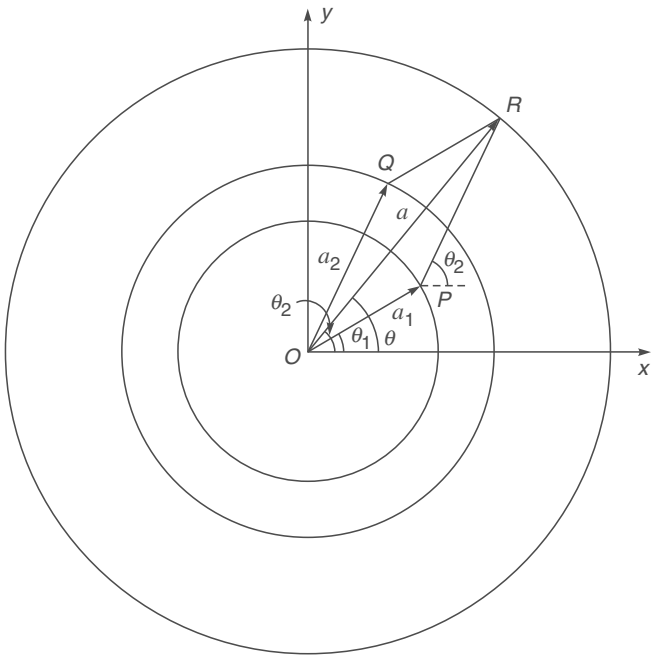


Fig. 13.8 The graphical method for determining the resultant of two simple harmonic motions along the same direction and having the same frequency.

consistent with Eqs. (13.23) and (13.24). Further, as the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ rotate on the circumference of the circles of radii a_1 and a_2, the vector $\overrightarrow{OR}$ rotates on the circumference of the circle of radius OR with the same frequency.

Thus, if we wish to find the resultant of the two displacements given by Eq. (13.20) then we must first draw a vector ($\overrightarrow{OP}$) of length a_1 making an angle θ_1 with the axis; from the tip of this vector we must draw another vector ($\overrightarrow{PR}$) of length a_2 making an angle θ_2 with the axis. The length of the vector $\overrightarrow{OR}$ will represent the resultant amplitude and the angle that it makes with the axis will represent the initial phase of the resultant displacement. It can be easily seen that if we have a third displacement

$$x_3 = a_3 \cos (\omega t + \theta_3) \qquad \textbf{(13.37)}$$

then from the point R we must draw a vector $\overrightarrow{RR'}$ of length a_3 which makes an angle θ_3 with the axis; the vector $\overrightarrow{OR'}$ will represent the resultant of x_1, x_2 and x_3.

As an illustration of the above procedure we consider the resultant of N simple harmonic motions all having the same amplitude and with their phases increasing in arithmetic progression. Thus,

$$x_1 = a \cos \omega t$$
$$x_2 = a \cos [\omega t + \theta_0] \qquad \textbf{(13.38)}$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$x_N = a \cos [\omega t + (N-1)\theta_0]$$

In Fig. 13.9, the vectors $\overrightarrow{OP_1}$, $\overrightarrow{P_1P_2}$, $\overrightarrow{P_2P_3}$, ..., correspond to x_1, x_2, x_3,..., respectively. The resultant is denoted by the

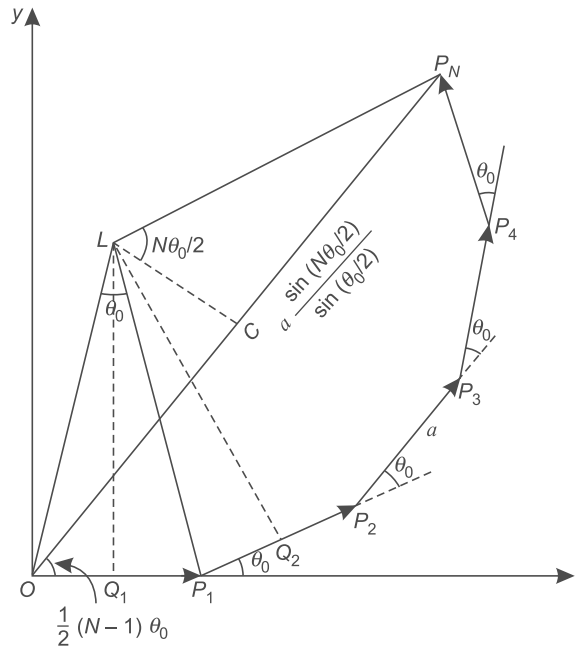


Fig. 13.9 The graphical method for determining the resultant of N simple harmonic motions along the same direction and having the same frequency.

* Clearly, if we assume the vector $\overrightarrow{OP}$ to rotate (in the anticlockwise direction) with angular velocity ω then the x-coordinate of the vector $\overrightarrow{OP}$ will be $a_1 \cos (\omega t + \theta_1)$ where $t = 0$ corresponds to the instant when the rotating vector is at the point P.